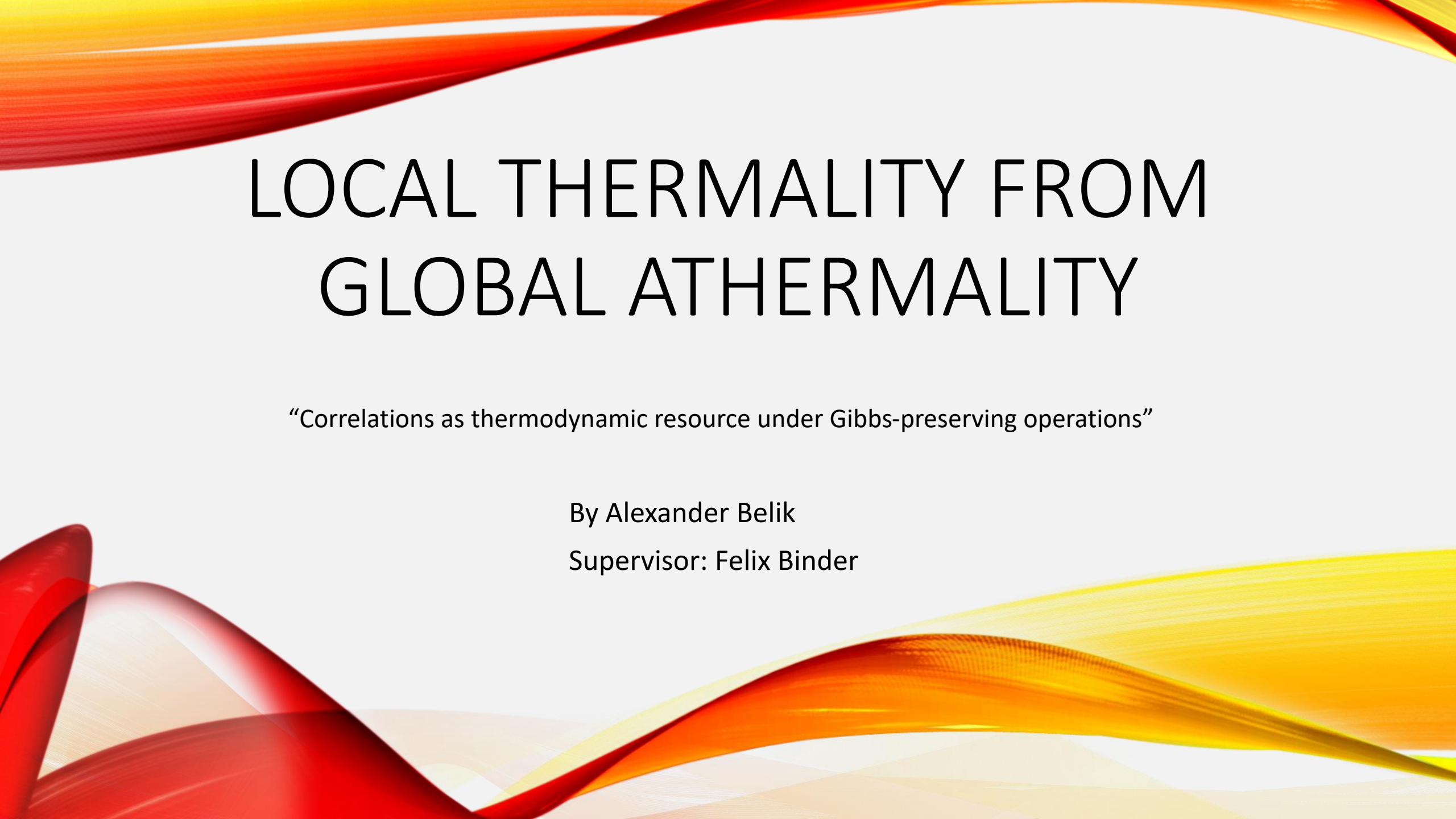




LOCAL THERMALITY FROM GLOBAL ATHERMALITY

“Correlations as thermodynamic resource under Gibbs-preserving operations”

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OBJECTIVE 1

1. Identify and characterise the set of bipartite states that are locally thermal (but not globally thermal).
2. Construct a hierarchy of these states, i.e. we say $\rho_{AB} \geq \sigma_{AB}$ if there exists a transformation $G_{AB}(\rho_{AB}) = \sigma_{AB}$ where G_{AB} is a Gibbs-preserving operation acting on the bipartite system AB .
3. Investigate how this hierarchy changes when the allowed operations are restricted to local Gibbs-preserving operations.



MOTIVATION: WHY CORRELATIONS MATTER THERMODYNAMICALLY

- Information $\leftrightarrow$ thermodynamics
- Correlations can store usable 'resource'
- Goal: Classify what you can do with that resource

BASICS: THERMAL EQUILIBRIUM STATE FOR ONE SYSTEM

Hamiltonian H , temperature $1/\beta$

Gibbs state: $\gamma = \frac{e^{-\beta H}}{Z}$, the “free” reference

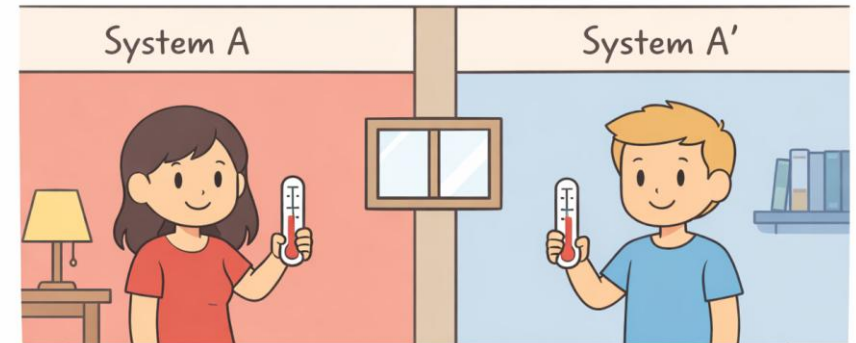
Interpretation: “no athermality relative to equilibrium”

TWO SYSTEMS AND SYMMETRY CHOICE

Two systems A and A'

Symmetric: $H_A = H_{A'}$, same β
 $\gamma_A = \gamma_{A'} = \gamma$

Reference joint equilibrium: $\gamma \otimes \gamma$



LOCALLY THERMAL STATES

Bipartite state:

$$\rho_{AA'} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_{A'})$$

Definition: $\text{Tr}_{A'}(\rho) = \gamma$, $\text{Tr}_A(\rho) = \gamma$

Locally looks thermal, globally may not



WHAT “EXTRA STRUCTURE” IS LEFT IF MARGINALS ARE FIXED?

All freedom is in correlations/coherence

$$\rho = \gamma \otimes \gamma + \mathcal{C}$$

Think: joint state = (equilibrium product) + (correlation piece)

$\mathcal{C}$ has zero marginals $\rightarrow$ doesn't change local states

QUANTIFYING THE RESOURCE

Key identity on LT: $D(\rho \parallel \gamma \otimes \gamma) = I(A : A')$

Relative entropy to $\gamma \otimes \gamma$:

$D(\rho \parallel \gamma \otimes \gamma)$ = “how far you are from equilibrium”.

Mutual information $I(A : A')$ = “how correlated the two systems are”.

Athermality = Correlations.

OBJECTIVE 2

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RESOURCE CAN'T INCREASE

Define allowed operations: “Gibbs-preserving” $G(\gamma \otimes \gamma) = \gamma \otimes \gamma$

Monotonicity = Equilibrium state unchanged

Data-processing Inequality

Claim: $D(\rho \parallel \gamma \otimes \gamma) \geq D(G(\rho) \parallel \gamma \otimes \gamma)$

Distinguishability can't increase under a noisy channel

RESOURCE CAN'T INCREASE II

Claim: $D(\rho \parallel \gamma \otimes \gamma) \geq D(\mathcal{G}(\rho) \parallel \gamma \otimes \gamma).$

For any quantum channel $\mathcal{G}$:

$$D(\rho \parallel \sigma) \geq D(\mathcal{G}(\rho) \parallel \mathcal{G}(\sigma)).$$

Take

$$\sigma = \gamma \otimes \gamma$$

Since

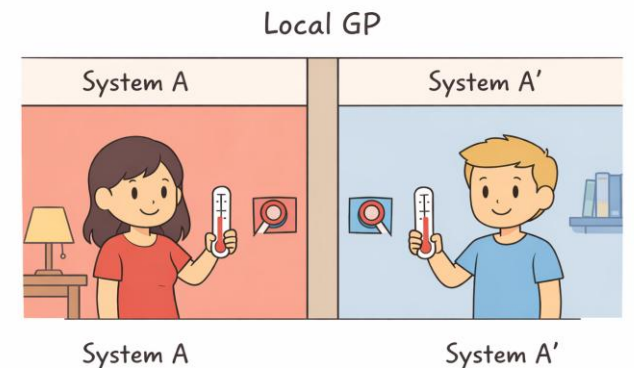
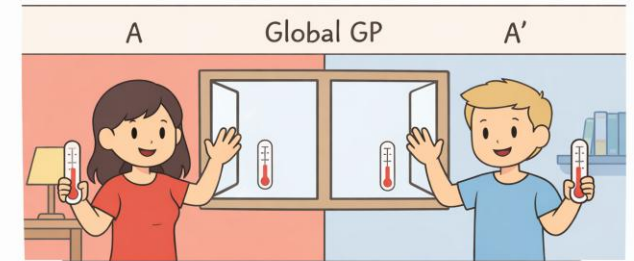
$$\mathcal{G}(\gamma \otimes \gamma) = \gamma \otimes \gamma,$$

We get:

$$D(\rho \parallel \gamma \otimes \gamma) \geq D(\mathcal{G}(\rho) \parallel \gamma \otimes \gamma).$$

GLOBAL VS LOCAL

- What changes when operations can't coordinate across the split?
- Global GP: any joint channel
 $G_{AA'}(\gamma \otimes \gamma) = \gamma \otimes \gamma$ preserving $\gamma \otimes \gamma$
- Local GP: $G_A \otimes G_{A'}$ with each preserving γ ,
 $G_X(\gamma) = \gamma$

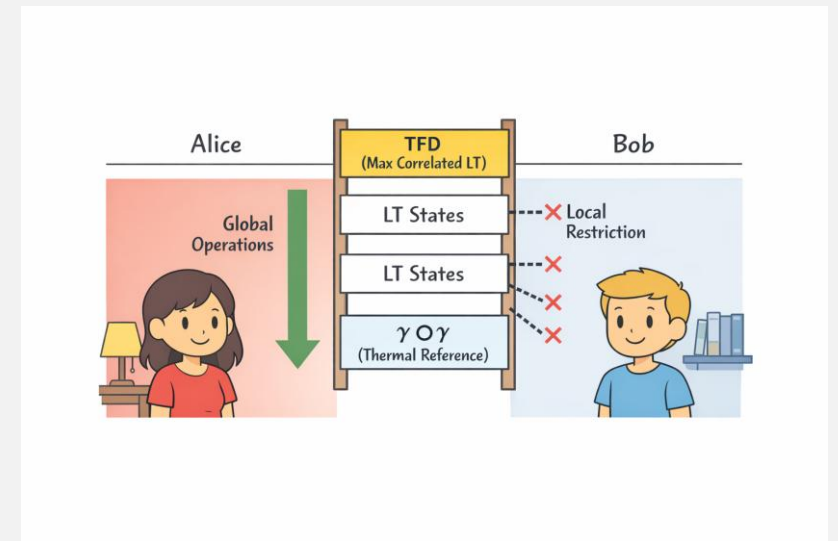


WHY TFD IS THE NATURAL TOP REFERENCE POINT?

- Thermo Field Double (TFD) state:

$$|\tilde{\gamma}\rangle = \sum_i \sqrt{g_i} |E_i\rangle_A |E_i\rangle_{A'}$$

- TFD is the Maximally correlated **pure** LT state.
- Anchor point for hierarchy.



OBJECTIVE 3

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RESOURCE ORDERING UNDER ALLOWED OPERATIONS

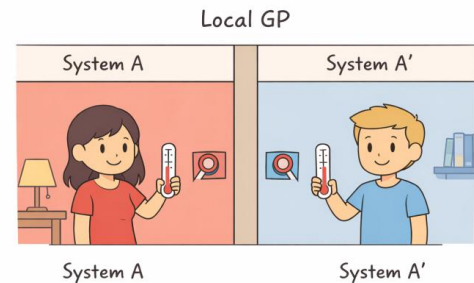
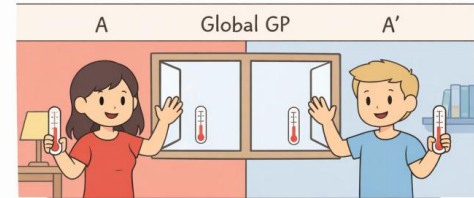
- Global

$\rho \rightarrow \sigma$ if $\exists G$ with $G(\gamma \otimes \gamma) = \gamma \otimes \gamma$ and $G(\rho) = \sigma$

- Local

$\rho \rightarrow \sigma$ if $\exists G_A, G_{A'}$ with $(G_A \otimes G_{A'}) (\rho) = \sigma$

How does this order change when restricting?
 $GP \rightarrow \text{Local } GP?$



HOW DO WE TEST ALLOWED OPERATIONS?

We represent a quantum channel using its Choi matrix J_G .

Convertibility = Semidefinite Program (SDP)

Physicality Constraints:

- $J_G \geq 0$ (*Complete Positivity*)
- $\text{Tr}_{out}(J_G) = I_{in}$ (*Trace Preservation*)

Convertibility Condition

- $\mathcal{G}(\rho) = \sigma$

Gibbs constraint

- $\mathcal{G}(\gamma \otimes \gamma) = \gamma \otimes \gamma$

The background features abstract, flowing waves in shades of red, orange, and yellow, creating a sense of movement and energy. The waves are layered, with some appearing more prominent than others, and they curve across the frame.

RESULTS

RESULTS – WHAT IS THE GEOMETRY?

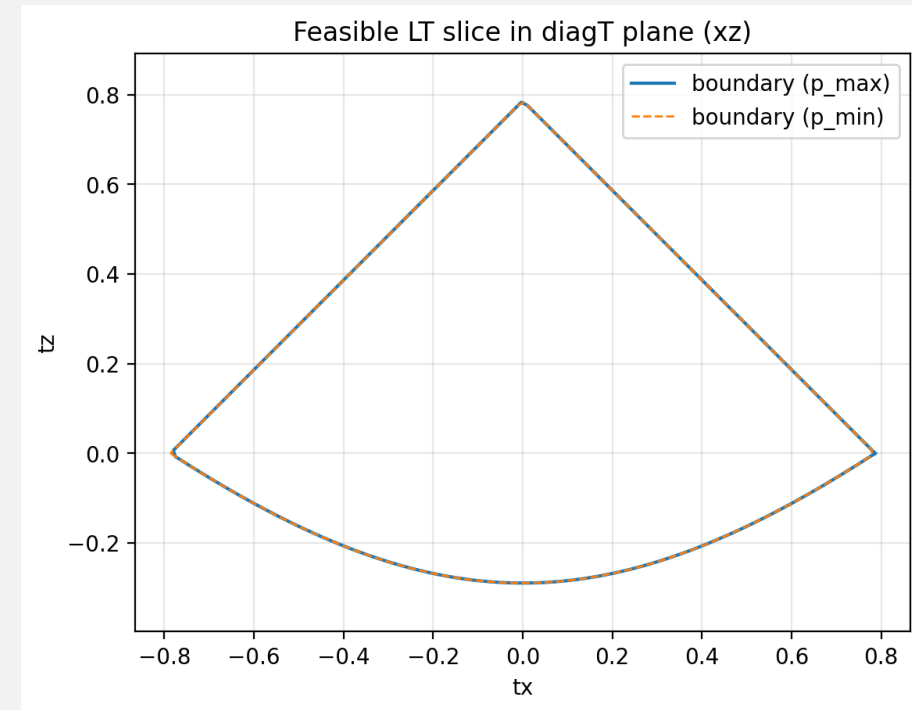
What does the locally thermal set actually look like geometrically?

The key geometric statement:

$$\text{LT} = \{\gamma \otimes \gamma + C \mid \text{linear constraints on } C \text{ } \rho \geq 0\}$$

Result

LT is a spectrahedron: a convex linear slice of the PSD cone.



RESULTS 2 - CORRELATIONS

How can we parameterise all locally thermal correlations?

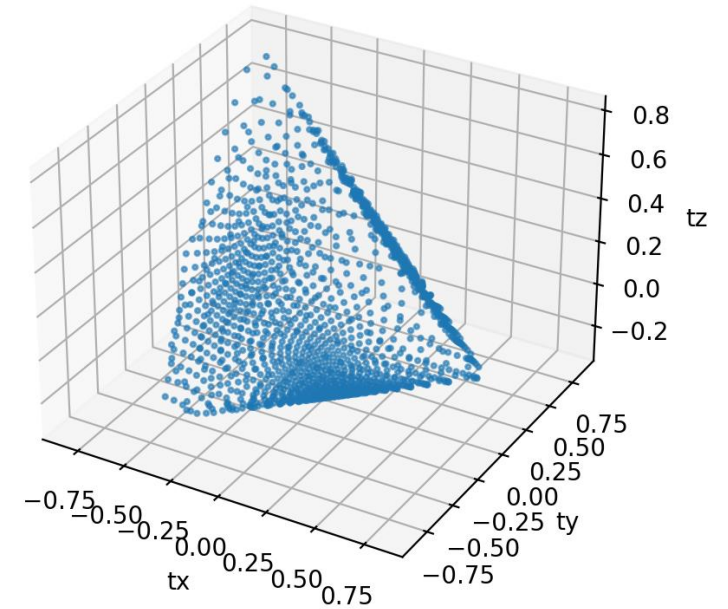
Correlation expansion

$$C = \frac{1}{4} \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j$$

$$\rho = \gamma \otimes \gamma + \frac{1}{4} \sum_{i,j} T_{ij} \sigma_i \otimes \sigma_j$$

$$T \in \mathbb{R}^{3 \times 3}$$

Feasible LT region boundary (diag-T slice, p_max surface)



RESULTS 3 – GLOBAL CONVERTIBILITY

Does mutual information fully determine convertibility under Global GP?

Mutual information

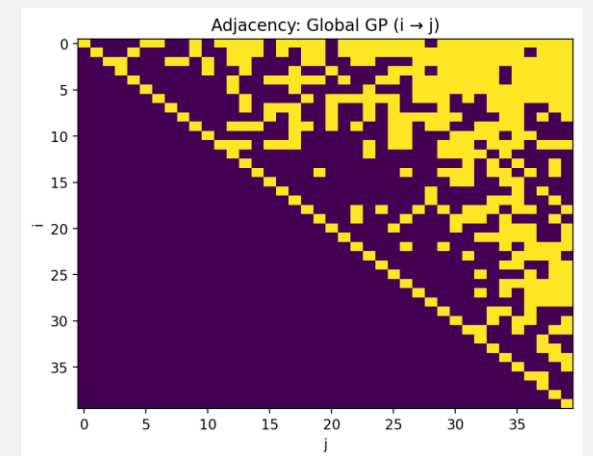
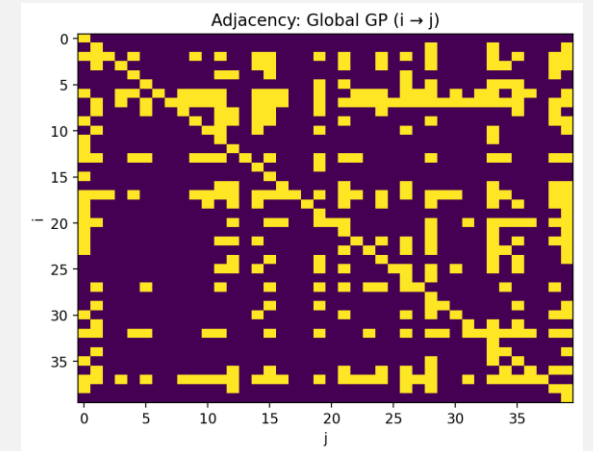
If we sort states by “how linked Alice and Bob are”.

One-way transformations

The convertibility plot becomes almost upper-triangular.

Theorem

$$\begin{aligned} D(\rho \parallel \gamma \otimes \gamma) &\geq D(\mathcal{G}(\rho) \parallel \gamma \otimes \gamma) \\ \Leftrightarrow \mathcal{G}(\gamma \otimes \gamma) &= \gamma \otimes \gamma \end{aligned}$$



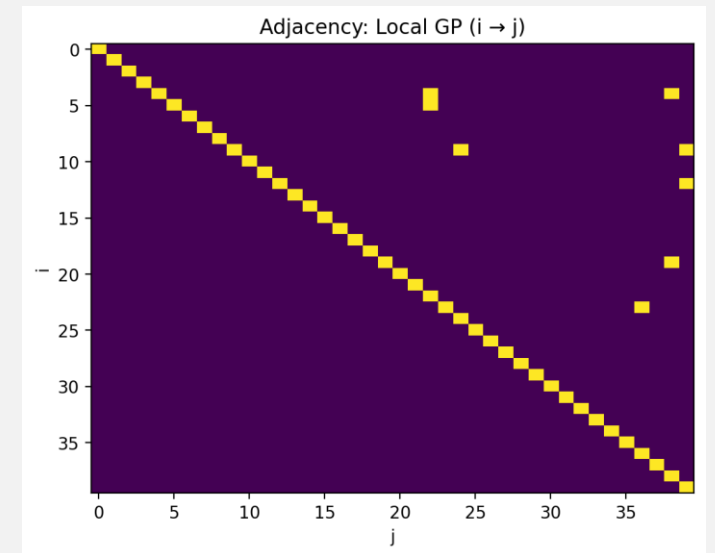
RESULTS 4 – LOCAL GP

Does the same scalar monotone govern Local GP?

Local GP = Alice and Bob each do their own actions,
no coordination.

Restricting operations changes the geometry of the
resource theory. **Local** $\subset$ **Global**.

$$I(\rho_i) \geq I(\rho_j) \Rightarrow \rho_i \rightarrow \rho_j \text{ under local GP}$$

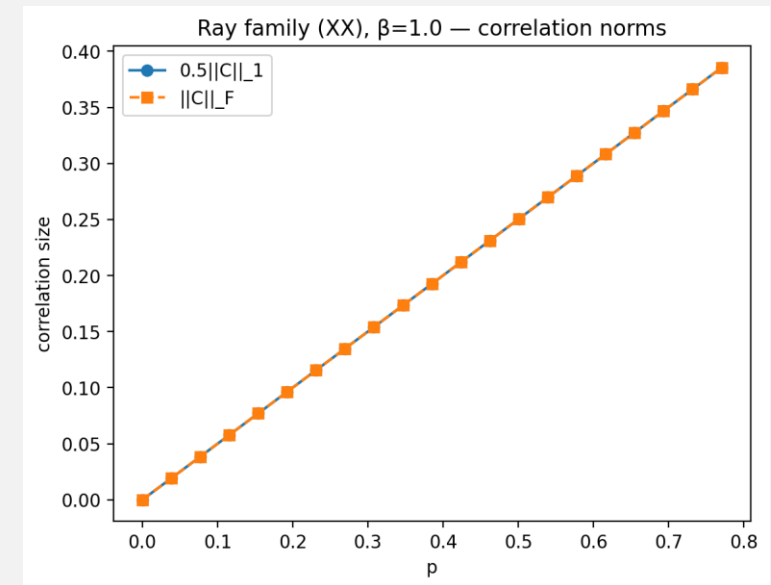


RESULTS 5 – RAY PROJECTION

Is local incomparability a high-dimensional effect vs 1D restriction?

Test a **1-parameter** LT “ray”

Define direction: $C_0 = \frac{1}{4} (X \otimes X)$



RESULTS 5 – RAY PROJECTION 2

Then define family:

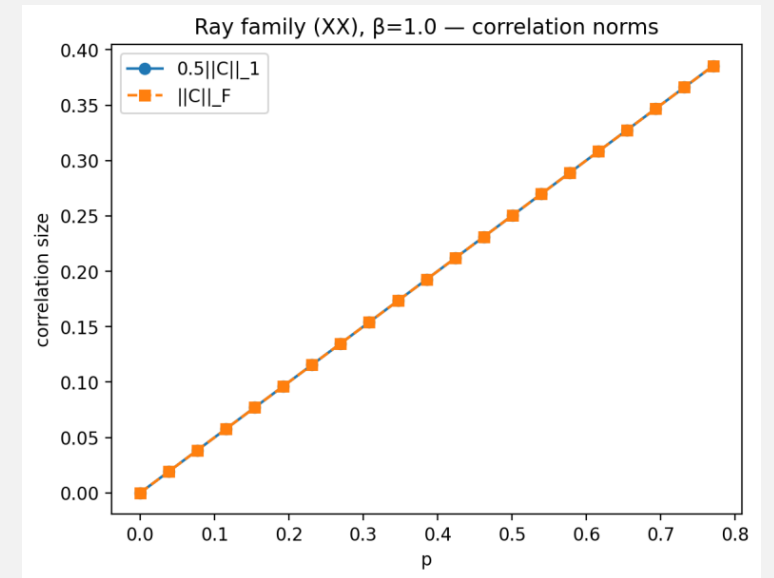
$$\rho(p) = \gamma \otimes \gamma + pC_0$$

Along this ray, local GP behaves like a **clean 1D chain**.

Then analytic PSD window:

$$p \in \left[-\frac{1}{\lambda_{\max}(C^{\sim})}, -\frac{1}{\lambda_{\min}(C^{\sim})} \right]$$

$$C^{\sim} = (\gamma^{-1/2} \otimes \gamma^{-1/2})C_0(\gamma^{-1/2} \otimes \gamma^{-1/2})$$



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BEYOND SLICES: THE FULL 9D LT GEOMETRY

Structure of the LT Set

$$\rho = \gamma \otimes \gamma + \frac{1}{4} \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j$$
$$\text{LT} = \{ T \in \mathbb{R}^{3 \times 3} \mid \rho(T) \geq 0 \}$$

What We've Seen So Far

- Diagonal slices
- 1D rays

What Remains

- Full 9-dimensional spectrahedron
- Coupled tensor constraints

WHY DOES INCOMPARABILITY OCCUR?

Observation

- Under verified Local GP, convertibility becomes extremely sparse in higher-dimensional correlation families.
- Along a fixed 1D ray, Local GP behaves like a monotone chain.

Current Investigation

What structure of the correlation matrix T is preserved under Local GP?

Exploring:

- Singular-value structure of T
- Tensor invariants under local CPTP maps
- SDP-based characterisation of locally reachable sets

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Q&A

Thank you